Supplementary material

Mingyang Li[1], Jiabei Liu[2,*], Jiayue Gao[2]

[1]Beijing Institute of Technology, Beijing, China

[2]Jinan University–University of Birmingham Joint Institute,
Jinan University, Guangzhou, China

[*]Corresponding author: Jiabei Liu
jxl2043@student.bham.ac.uk

2 November 2026

Model selection and performance estimation can diverge: a RoadSens-4M audit and a nine-acquisition PVS case study

Working version 9, 2 November 2026. This supplement accompanies `main_body.tex`. All tables refer to completed canonical runs unless expressly labelled exploratory. The RoadSens main tables report five-fold arithmetic means; PVS tables specify three outer-domain means or individual domains; tables explicitly labelled pooled OOF use the combined heldout predictions. Computational fit-fold counts are not independent scientific sample sizes. The original 1,580-fit suite and targeted post-draft extensions are reported separately.

S1. Execution matrix and evidence index

Block	Fixed comparison	New canonical fit-folds	Authoritative output
A	Four models × four feature sets × three splits × two policies × five folds; seed 42	480	`../runs/main_A_v2/analysis/main_A_analysis.md`
R	RF seeds 43/44 × four features × random/recording × two policies × five folds	160	`../runs/consistent_runtime_R_v1/`; `../runs/auxiliary_summary_v1.md`
B	Three non-RF models × full/context × random/recording × two policies × seeds 43/44 × five folds	240	`../runs/phase_B_v1/`; `../runs/auxiliary_summary_v1.md`
C	RF × eight ablations × three splits × two policies × five folds	240	`../runs/phase_C_v2/independent_validation.md`
D	Three additional capacity/budget combinations × three splits × two policies × five folds	90	`../runs/phase_D_capacity_vs_A_v1.md`
E	Binary RF × four feature sets × three splits × five folds	60	`../runs/phase_E_v1/`; `../runs/auxiliary_summary_v1.md`
H	Specified component-heldout H1 (100) and byte-copy-removal H2 (90) fits	190	`../runs/content_overlap_v2/analysis/`
I	Two spatial grids × two models × three feature groups × five folds	60	`../runs/spatial_binary_v1/independent_summary.md`
S	STRIDE: two models × two feature sets × three seeds × five folds	60	`../runs/stride_supplement_v2/validation_summary.md`
Total	Includes120 A RF reproduction fits; excludes330 earlier exploratory fits	**1,580**	`../execution_inventory.json`

All canonical blocks completed without a recorded fit failure and underwent independent source/prediction/metric checks. The shared environment is Python3.13.12, scikit-learn1.9.0, NumPy2.5.2, pandas3.0.5, joblib1.5.3, SciPy1.18.0, and threadpoolctl3.6.0. These CPU experiments use at most four threads per fit; a GPU is not required. Hardware timings are not controlled algorithmic speed comparisons.

The immutable cache hash is `fa2078df1857f7872c13c6fe187f0883d85a23f9d15fdb3c6897a53314fdea16`. The parent pilot, shared runner, auxiliary driver, source archives, exact split maps, preflights, and model hashes are separately indexed in the execution inventory. Read `../reproducibilitynotes` for commands rather than reconstructing splits from seed values.

S2. Contribution and prior-work boundary

Topic	Established or closest source	Specific evidence added here	Claim excluded
Smartphone road classification	RoadSens data descriptor and archive; CKDP-PINNFed	A fixed v3 payload reconstructed into explicit cohorts and comparable fit/test membership	A first RoadSens classifier or exact replication of the source baseline
Contextual/GPS information	RoadSens source fields; CKDP-PINNFed coordinate processing	GPS/availability/context controls under two declared blank-label policies	First GPS use; GPS is always harmful; mask performance proves causal leakage
Structured evaluation	Roberts et al.; Ploton et al.	File, content-component, and buffered spatial diagnostics on this snapshot	A new cross-validation algorithm or a verified independent-road benchmark
Data redundancy	This version's exact file and finite-axis audit	A byte-identical pair and three exact-content connected groups, with non-monotone predictive changes after regrouping	All score inflation is caused by duplicates; matching content proves route identity
Separate-source data	STRIDE descriptor and v4 archive	Seventeen offline recording-type examples with duration controls	Full three-class external validation, transfer, or new-device generalization

The Chinese contribution matrix supplies more detailed claim-to-evidence links. The reference-verification note distinguishes source statements from locally observed results and records access limitations. This is a bounded review, not a priority certificate.

S3. Same-runtime RF seed checks

Table S1: Full-numeric random-minus-recording differences of five-fold mean macro-F1.

Policy	Seed 42	Seed 43	Seed 44
Primary	0.5656	0.5387	0.5354
Blank-as-normal	0.5360	0.5548	0.5148

Each of the five fold-index differences has the same positive direction within every seed/policy configuration. Fold numbers from different splitting schemes do not represent matched independent roads; no paired-road significance test is inferred. Additional full/context model results for all specified B configurations, including weaker logistic-regression patterns, appear in the final auxiliary summary. No best seed or feature combination is selected for the primary table.

Historical seed 43/44 serialized RF estimators used sklearn1.5.2, while seed 42 used1.9.0. They remain historical evidence; R refits the frozen source/row partitions under1.9.0 and supplies the same-runtime comparison above. Original source-level assignments are preserved because regenerating a stratified group partition under a different library version need not reproduce it.

S4. All main conditional uncertainty contrasts

All estimates below are **pooled OOF macro-F1 differences**, not fold-mean differences. Components, strata, and the 2,000 shared weight vectors are frozen. Intervals use the 2.5th and 97.5th empirical percentiles and retain fixed

fitted predictors. Every listed contrast has2,000/2,000 valid draws. Strata contain only five normal-only components and 91 mixed components; these intervals cannot establish a general road/device population or training uncertainty.

Table S2: Twenty-four specified main-model contrasts.

Policy	Model	Contrast	Estimate	Conditional95% interval
Primary	RF100	Full random − recording	+0.5827	[+0.4363,+0.6960]
Primary	RF100	Full random − temporal	+0.1798	[+0.1284,+0.2398]
Primary	RF100	Recording context − full	+0.1853	[+0.0812,+0.3154]
Primary	Extra Trees	Full random − recording	+0.5480	[+0.3917,+0.6679]
Primary	Extra Trees	Full random − temporal	+0.1973	[+0.1411,+0.2593]
Primary	Extra Trees	Recording context − full	+0.0047	[−0.1596,+0.1781]
Primary	HistGB	Full random − recording	+0.5249	[+0.3928,+0.6350]
Primary	HistGB	Full random − temporal	+0.1490	[+0.1099,+0.1991]
Primary	HistGB	Recording context − full	+0.1015	[−0.0648,+0.3390]
Primary	Logistic regression	Full random − recording	+0.3411	[+0.2129,+0.4169]
Primary	Logistic regression	Full random − temporal	+0.0424	[+0.0186,+0.0690]
Primary	Logistic regression	Recording context − full	−0.1297	[−0.2353,−0.0596]
Blank-as-normal	RF100	Full random − recording	+0.5455	[+0.4881,+0.6042]
Blank-as-normal	RF100	Full random − temporal	+0.3496	[+0.2820,+0.4257]
Blank-as-normal	RF100	Recording context − full	−0.0364	[−0.0893,+0.0164]
Blank-as-normal	Extra Trees	Full random − recording	+0.4760	[+0.4232,+0.5338]
Blank-as-normal	Extra Trees	Full random − temporal	+0.3346	[+0.2712,+0.4002]
Blank-as-normal	Extra Trees	Recording context − full	−0.0360	[−0.0809,+0.0072]
Blank-as-normal	HistGB	Full random − recording	+0.5303	[+0.4666,+0.5905]
Blank-as-normal	HistGB	Full random − temporal	+0.3107	[+0.2575,+0.3718]
Blank-as-normal	HistGB	Recording context − full	+0.0247	[−0.0119,+0.0666]
Blank-as-normal	Logistic regression	Full random − recording	+0.1197	[+0.0578,+0.1822]
Blank-as-normal	Logistic regression	Full random − temporal	+0.0444	[+0.0241,+0.0644]
Blank-as-normal	Logistic regression	Recording context − full	−0.0382	[−0.0852,+0.0071]

Table S3: Additional paired RF ablations using the same draws.

Policy	Contrast	Estimate	Conditional95% interval
Primary	Recording numeric-minus-context − full	−0.1179	[−0.2486,−0.0208]
Blank-as-normal	Recording numeric-minus-context − full	−0.0610	[−0.1064,−0.0185]

The prediction row IDs, truths, and stored estimator class order are aligned before comparing A with C. The C validation file stores these paired results; the main A bootstrap directory stores component confusions, draws, and provenance hashes. These are predictive comparisons after fitting different feature representations, not causal interventions on acquisition context.

S5. Content-component and byte-copy-removal designs

H1 uses the same 101-source supervised cohort as the main analysis. Seed 42 fits RF100 and HistGB with full/context/IMU features under both policies (60 fit-folds). Seeds 43/44 fit RF100 with full/context features under both policies on their own component-heldout maps (40). Total H1 is 100. H2 excludes source 47 by the label-blind

duplicate-file rule and fits RF100 for full/context/IMU, random/recording/temporal, both policies (90). Both H1 and H2 preserve their complete frozen cohorts and report every specified configuration.

For H1 seed 42 versus main recording holdout, twelve same-cohort pooled comparisons comprise six increases and six decreases. The range approximately−0.0462 to+0.0340 is not a duplicate-effect confidence interval. Group changes alter training/testing composition. H2 removes only a byte copy and leaves the partial inertial overlaps; it should not be labelled completely deduplicated.

Table S4: H2 RF full-feature pooled OOF macro-F1, rounded to three decimals.

Policy	Cohort supervised rows	Random	Temporal	Recording
Primary	32,838	0.995	0.818	0.445
Blank-as-normal	61,484	0.931	0.583	0.391

These values are not paired effects against the 101-file main table. Ten policy/fold H2 time-block membership checks verify retention of the original inclusive one-second embargo. The H independent analysis checks all 190 model hashes and 1,795,048 heldout prediction rows across configurations; this prediction-row count repeatedly includes the same underlying observations and is not a sample-size claim.

S6. STRIDE recording-level supplement

Table S5: Pooled OOF macro-F1 across the same 17 STRIDE observations.

Features	Model	Seed 42	Seed 43	Seed 44
Ten sensor norm summaries	RF100	0.7639	0.8235	0.8235
Ten sensor norm summaries	Scaled logistic regression	0.7639	0.8786	0.8211
Duration only	RF100	0.5882	0.5278	0.4706
Duration only	Scaled logistic regression	0.3561	0.4396	0.3529

The two sensor streams are summarized separately; asynchronous timestamps are not interpolated into invented event windows. A recording folder provides one whole-record type label. Duration is deliberately separated from sensor summaries. Small and stratified test folds, a single device identifier, lack of Normal labels, and unverified physical-route independence prevent deployment or external-three-class conclusions.

The official v4 ZIP has SHA-256 `475853a7aaf9a15068dc9c17014f3e46197acc4b705cd3851b84692cdd82f851`. The official source record reports CC BY4.0; author attribution and any applicable third-party requirements still belong in a public release plan. No original data are publicly redistributed by this working manuscript.

S7. Exploratory temporal controls and annotation runs

Before the expansion, RF primary/sensitivity recording-heldout macro-F1 for a raw-axis one-second statistics control was0.3953/0.3972. The twenty-dimensional physical one-second representation scored0.3751/0.3759, and its prefix-calibrated variant0.4015/0.3706. These historical scores informed closing the proposed physical-representation branch. However, targeted runtime verification subsequently identifies sklearn1.5.2 in the historical raw-axis control, whereas the main baseline environment is 1.9.0. The old comparison is therefore cross-version exploratory evidence, not a controlled same-runtime representation ranking. New same-runtime window results

must determine any updated claim; no new physical-method contribution is inferred from the old scores. These fits are not additional canonical expansion fits.

The raw-axis control has60 dimensions versus20 for the physical representation. These different capacities do not prove one is inferior in all settings. Rolling windows calculated before row splitting can also share raw signal context across heldout rows. Consequently, temporal-feature random-row results are not independent-window evidence. Main expansion features are instantaneous per-row representations and are not described as event or sequence models.

The 348 continuous labelled runs are149 Pothole plus199 Bump runs. All have sampled rows. Secondary equal-run-weight row agreement and majority-label correctness describe these observed segments; neither validates physical boundaries, new-event recall, localization delay, or false alarms per hour. Details remain in `../annotated_runs_README.md` and the accompanying machine-readable run summaries.

S8. Retained failures, revisions, and publication boundary

Main_A_v1 stopped at an RF reproduction mismatch under the unintended sklearn1.5.2 interpreter. Its diagnostic and failed attempt are retained. A correct-runtime diagnostic reproduces6,577/6,577 labels for the checked first fold, and all 120 main A RF reproduction folds subsequently pass. STRIDE_v1 failed before fitting because of a manifest NameError; v2 is canonical. The unexecuted first H implementation omitted the temporal embargo and was corrected before H_v2 fitting. Earlier content-map drafts, checks and version-specific corrections are retained. C_v1 was interrupted during scheduler handoff; full C_v2 is canonical and the partial run is excluded from the total.

These revisions do not represent score-based selection of favourable fits. Incorrect/partial versions do not fill planned matrix cells. The reproducibility guide identifies commands and authoritative outputs. The public code, protocols, compact evidence and manuscript are available at `https://github.com/Asuna-art/roadsens-evaluation-audit`. The initial public code snapshot is commit `66eb1995a61460c866581122fa86ea908d08b640`. Public views omit signal matrices, direct label/prediction arrays and coordinates and do not replace complete historical inputs. Raw data and model binaries are excluded. The repository README states the environment and replay limitations; author-contribution and venue-specific declarations still require completion.

S9. Targeted anchor-stride sensitivity

The frozen post-draft design is recorded in `../key_experiments/protocol_key_v1.md`. Only strides 5 and 20 add new RF100 fits; stride 10 is the original A reference. Both label policies, full/IMU features and the original filename fold mapping are retained. New stride populations are reconstructed from full-cadence arrays and live raw annotations. Before fitting, reconstruction at stride 10 matches the cached full/IMU feature values, ordered labels and source identities, and all original fit/test sequences. All 101 source files match their frozen hashes.

Table S6: All eight new sampling configurations; recording-heldout seed 42, RF100.

Stride	Policy	Features	Rows	Fold mean macro-F1	Pooled OOF macro-F1
5	Primary	Full numeric	65,774	0.4386	0.4323
5	Primary	IMU12	65,774	0.3379	0.3408
5	Blank-as-normal	Full numeric	123,411	0.3866	0.3726
5	Blank-as-normal	IMU12	123,411	0.3432	0.3402
20	Primary	Full numeric	16,440	0.4086	0.4049
20	Primary	IMU12	16,440	0.3386	0.3400
20	Blank-as-normal	Full numeric	30,889	0.3765	0.3648
20	Blank-as-normal	IMU12	30,889	0.3563	0.3492

The source count remains101 at every stride, and allfolds retain the fixed three classes. Independently recomputed predictions from all 40 reloaded models match the saved OOF predictions; source-row/truth/fold/model hashes also pass. Different strides alter the observed and capped fitting cohorts, so these are descriptive density sensitivities, not same-example paired effects or new independent acquisitions. No random-row experiment is repeated here, and no across-stride random-minus-recording claim is made.

S10. Verified temporal controls and runtime amendment

The independent preflight certifies the original stride-ten populations and fixed file/timeblock splits. Window inputs are raw source rows in the closed past-inclusive one-second interval, with finite timestamps and stable source-row ordering for ties. Membership agrees with a naive reference in 13 synthetic cases and 303 real anchors from 101 sources. All 20 policy/split/fold membership arrays and deterministic fit caps pass. For the ten timeblock policy-folds, the original anchor embargo already gives zero raw-window intersections; no additional rows are removed. This does not remove content overlap between different source files.

Table S7: All 16 temporal configurations. Fold means and pooled OOF are distinct summaries of the same predictions.

Policy	Split	Model	Features	Fold mean macro-F1	Pooled OOF macro-F1
Primary	Recording	RF100	IMU12	0.3410	0.3423
Primary	Recording	RF100	Window60	0.4072	0.4042
Primary	Recording	HistGB	IMU12	0.3379	0.3356
Primary	Recording	HistGB	Window60	0.4173	0.4161
Primary	Time-block	RF100	IMU12	0.4454	0.4498
Primary	Time-block	RF100	Window60	0.5929	0.5970
Primary	Time-block	HistGB	IMU12	0.4419	0.4458
Primary	Time-block	HistGB	Window60	0.6015	0.6054
Blank-as-normal	Recording	RF100	IMU12	0.3448	0.3422
Blank-as-normal	Recording	RF100	Window60	0.4100	0.4039
Blank-as-normal	Recording	HistGB	IMU12	0.3392	0.3373
Blank-as-normal	Recording	HistGB	Window60	0.4023	0.3996
Blank-as-normal	Time-block	RF100	IMU12	0.3591	0.3614
Blank-as-normal	Time-block	RF100	Window60	0.4256	0.4280
Blank-as-normal	Time-block	HistGB	IMU12	0.3581	0.3603
Blank-as-normal	Time-block	HistGB	Window60	0.4294	0.4317

There are eight matched feature comparisons: four recording and four timeblock combinations of policy and classifier.

Their fold-mean and pooled summaries are not additional independent replications. All 80 T models are reloaded to reproduce their saved predictions and metrics; source rows, true labels, explicit integer/string encodings, class supports/recalls, preprocessing, and model/prediction hashes are checked. Every configuration tests each expected supervised source-row exactly once. The 40 instantaneous controls additionally reproduce A after row-ID alignment and explicit class decoding.

S10.1. Same-runtime reference and retained failed attempts

All ten historical axis-window models were serialized with sklearn1.5.2. Their parameters, retained-feature indices, and training-median arrays agree with the inspected frozen definitions, but their predictions do not provide a same-runtime equality target for1.9.0. This is a verified provenance discrepancy; the checked identities/imputation do not isolate a library change as the sole causal mechanism of every prediction difference.

The unchanged original axis-window driver is therefore rerun in the locked 1.9.0 interpreter, yielding ten separately saved reference models. All ten reference model hashes, versions and parameters are checked. T's ten RF/window recording cells match these references on exact fit/test identities and every predicted label before the other 70 fixed cells are executed. The protocol amendment, code, model and source hashes are retained. No score threshold, new parameter choice or favourable-seed selection is introduced.

T_v1 fitted two models before a prediction-object-array read error stopped verification. T_v2 fitted two models before the prediction gate found the historical runtime mismatch. Both attempts are preserved and excluded from canonical completion. T_v3 uses Unicode prediction arrays and the same-runtime original-implementation reference; the scientific data/feature/split/model design remains unchanged. The original historical window results remain in Section S7 with their cross-version limitation.

S10.2. Completion and evidence inventory

Targeted block	Canonical fit-folds	Independent checks
Anchor stride 5/20 sampling	40	8 configurations, all 40 reloaded predictions and model hashes verified
Same-runtime original-window reference	10	All 10 serialized model versions, parameters and hashes checked; predictions reproduced by T
T temporal diagnostic	80	16 configurations; all 80 source/label/model/prediction checks pass;40 controls match A
Targeted total	130	Distinct from the original 1,580-fit suite
Combined canonical total	1,710	Computational fits, not independent acquisitions

The authoritative extension inventory is `../key_experiments/final_key_inventory_v1.json`; it references the unchanged original execution inventory by SHA. Primary T validation is `../key_experiments/T_temporal_v3/independent_validation_v1.json`, with readable aggregate tables in `independent_aggregate_v1.md`. Sampling validation is `../key_experiments/runs/S_sampling_v1/independent_validation.md`. Neither extension supplies new road-normal truth, independent normal-source acquisitions, verified route/device identifiers, or an external three-class benchmark.

S11. Independent-source PVS nested evaluation

The PVS source review, source-byte hashes, feature definitions and frozen protocols are in `external_pvs_v1`. Every acquisition uses only its left dashboard accel/gyro axes; its label table is joined by the authors' sample-index organization after exact length checks. Tables S8–S16 preserve both outer axes and both declared window grids, with class recalls and exact confusion counts in separate readable tables. No-speed-bump is not a general Normal label. PVS is not used for RoadSens-to-PVS classifier transfer or an aligned three-class test.

The primary grid contains 43,204 windows from nine acquisitions and 1,080,905 original rows; the sensitivity contains 10,804 disjoint windows from those same acquisitions. For each grid, 240 inner fits and 24 outer refits give 264 models. Selections are saved before outer predictions and remain immutable; separate joins report outer scores. The same four outer predictions are reused by the three validation selectors. Before this post hoc follow-up, there were 528 independently verified PVS fits and 2,238 fits in the combined computational inventory; the updated totals are 576 PVS fits and 2,286 fits overall. These fit-folds are not independent experiments or newly sampled roads.

The selected-system MAE definition compares the chosen candidate's own inner estimate with its outer score, then averages absolute errors across the three domains of one axis. Primary errors are scenario: 0.1696 to 0.1357; vehicle: 0.0530 to 0.0445; nonoverlap errors are scenario: 0.1069 to 0.1290; vehicle: 0.0489 to 0.0535. The primary analysis did not meet the declared two-axis criterion. The two-axis threshold is 0.02, prescribed before fitting. All candidate and selector outcomes, including contrary values, are retained below. Regret uses the best of the same four predeclared outer candidates as a descriptive oracle, never as a model-selection procedure. Training sizes and class supports accompany the exact maps; cap and sample-count differences limit causal interpretation.

Table S8: PVS source population by trip

Trip	Vehicle/driver	Scenario	Raw rows	Windows	Speed-bump windows
PVS 1	Volkswagen Saveiro / Driver 1	1	144036	5758	138
PVS 2	Volkswagen Saveiro / Driver 1	2	124684	4983	79
PVS 3	Volkswagen Saveiro / Driver 1	3	105816	4229	24
PVS 4	Fiat Bravo / Driver 2	1	132492	5296	124
PVS 5	Fiat Bravo / Driver 2	2	133877	5352	77
PVS 6	Fiat Bravo / Driver 2	3	96279	3848	25
PVS 7	Fiat Palio / Driver 3	1	128548	5138	116
PVS 8	Fiat Palio / Driver 3	2	123618	4941	74
PVS 9	Fiat Palio / Driver 3	3	91555	3659	25

Table S9: Primary PVS candidate metrics by vehicle/driver domain

Outer domain	Candidate	Random CV	Trip CV	Matched CV	Outer F1	Outer BA
vehicle_1	RF100 IMU	0.8639	0.8137	0.7974	0.8599	0.7983
vehicle_1	RF100 IMU+context	0.8972	0.8332	0.8124	0.8543	0.8001
vehicle_1	HGB IMU	0.8970	0.8391	0.8187	0.8641	0.8025
vehicle_1	HGB IMU+context	0.9217	0.8664	0.8618	0.8492	0.7858
vehicle_2	RF100 IMU	0.8670	0.8131	0.7356	0.8155	0.8499
vehicle_2	RF100 IMU+context	0.8763	0.8059	0.7726	0.8625	0.8823
vehicle_2	HGB IMU	0.8968	0.8397	0.7652	0.8494	0.8859
vehicle_2	HGB IMU+context	0.9144	0.8470	0.7985	0.9049	0.9164
vehicle_3	RF100 IMU	0.8731	0.8168	0.7884	0.7532	0.9253
vehicle_3	RF100 IMU+context	0.8964	0.8419	0.8349	0.7887	0.9262
vehicle_3	HGB IMU	0.9086	0.8500	0.8319	0.7917	0.9421
vehicle_3	HGB IMU+context	0.9268	0.8659	0.8352	0.8497	0.9459

Table S10: Primary PVS candidate metrics by scenario domain

Outer domain	Candidate	Random CV	Trip CV	Matched CV	Outer F1	Outer BA
scenario_1	RF100 IMU	0.9147	0.8757	0.8574	0.7186	0.6623
scenario_1	RF100 IMU+context	0.9248	0.8931	0.6568	0.6824	0.6246
scenario_1	HGB IMU	0.9268	0.9026	0.8474	0.7152	0.6536
scenario_1	HGB IMU+context	0.9461	0.9095	0.6701	0.6702	0.6129
scenario_2	RF100 IMU	0.8401	0.7758	0.7368	0.8781	0.9210
scenario_2	RF100 IMU+context	0.8744	0.8057	0.6806	0.6318	0.5920
scenario_2	HGB IMU	0.8891	0.8043	0.7513	0.9026	0.9241
scenario_2	HGB IMU+context	0.9280	0.8499	0.6851	0.7074	0.6470
scenario_3	RF100 IMU	0.8612	0.8209	0.7697	0.8866	0.8711
scenario_3	RF100 IMU+context	0.8823	0.8339	0.7466	0.8588	0.8973
scenario_3	HGB IMU	0.9018	0.8498	0.7464	0.8845	0.8510
scenario_3	HGB IMU+context	0.9265	0.8792	0.7201	0.9143	0.8782

Table S11: Primary PVS selector metrics by vehicle/driver domain

Outer domain	Selector	Selected candidate	Inner estimate	Outer F1	Absolute error	Oracle regret
vehicle_1	Random	HGB IMU+context	0.9217	0.8492	0.0725	0.0149
vehicle_1	Trip	HGB IMU+context	0.8664	0.8492	0.0172	0.0149
vehicle_1	Matched	HGB IMU+context	0.8618	0.8492	0.0126	0.0149
vehicle_2	Random	HGB IMU+context	0.9144	0.9049	0.0095	0.0000
vehicle_2	Trip	HGB IMU+context	0.8470	0.9049	0.0579	0.0000
vehicle_2	Matched	HGB IMU+context	0.7985	0.9049	0.1064	0.0000
vehicle_3	Random	HGB IMU+context	0.9268	0.8497	0.0771	0.0000
vehicle_3	Trip	HGB IMU+context	0.8659	0.8497	0.0162	0.0000
vehicle_3	Matched	HGB IMU+context	0.8352	0.8497	0.0145	0.0000

Table S12: Primary PVS selector metrics by scenario domain

Outer domain	Selector	Selected candidate	Inner estimate	Outer F1	Absolute error	Oracle regret
scenario_1	Random	HGB IMU+context	0.9461	0.6702	0.2760	0.0484
scenario_1	Trip	HGB IMU+context	0.9095	0.6702	0.2393	0.0484
scenario_1	Matched	RF100 IMU	0.8574	0.7186	0.1388	0.0000
scenario_2	Random	HGB IMU+context	0.9280	0.7074	0.2206	0.1951
scenario_2	Trip	HGB IMU+context	0.8499	0.7074	0.1424	0.1951
scenario_2	Matched	HGB IMU	0.7513	0.9026	0.1513	0.0000
scenario_3	Random	HGB IMU+context	0.9265	0.9143	0.0122	0.0000
scenario_3	Trip	HGB IMU+context	0.8792	0.9143	0.0351	0.0000
scenario_3	Matched	RF100 IMU	0.7697	0.8866	0.1169	0.0277

Table S13: Non-overlap sensitivity candidate performance by held-out axis

Axis	Candidate	Random CV	Trip CV	Matched CV	Outer F1	Outer BA
Scenario	HGB IMU	0.8791	0.8512	0.7926	0.8168	0.7914
Scenario	HGB IMU+context	0.8851	0.8561	0.6804	0.7986	0.7522
Scenario	RF100 IMU	0.8217	0.8006	0.7606	0.7872	0.8134
Scenario	RF100 IMU+context	0.8497	0.8293	0.7229	0.7862	0.7924
Vehicle	HGB IMU	0.8578	0.8301	0.7995	0.8282	0.8539
Vehicle	HGB IMU+context	0.8669	0.8446	0.8217	0.8485	0.8587
Vehicle	RF100 IMU	0.8040	0.7825	0.7538	0.7871	0.8611
Vehicle	RF100 IMU+context	0.8336	0.8073	0.7818	0.8151	0.8774

Table S14: Non-overlap sensitivity selector estimation summary

Axis	Selector	Mean estimate	Mean outer F1	MAE	Outer-minus-inner bias
Scenario	Trip	0.8608	0.7954	0.1061	-0.0655
Scenario	Matched	0.7926	0.8168	0.1290	0.0242
Scenario	Random	0.8869	0.7904	0.1069	-0.0965
Vehicle	Trip	0.8446	0.8485	0.0497	0.0038
Vehicle	Matched	0.8240	0.8522	0.0535	0.0282
Vehicle	Random	0.8672	0.8380	0.0489	-0.0291

Table S15: Primary PVS class recalls and confusion counts by vehicle/driver domain

Outer domain	Candidate	No-bump recall	Bump recall	TN	FP	FN	TP
vehicle_1	RF100 IMU	0.9991	0.5975	14716	13	97	144
vehicle_1	RF100 IMU+context	0.9986	0.6017	14708	21	96	145
vehicle_1	HGB IMU	0.9992	0.6058	14717	12	95	146
vehicle_1	HGB IMU+context	0.9990	0.5726	14715	14	103	138
vehicle_2	RF100 IMU	0.9919	0.7080	14154	116	66	160
vehicle_2	RF100 IMU+context	0.9946	0.7699	14193	77	52	174
vehicle_2	HGB IMU	0.9931	0.7788	14172	98	50	176
vehicle_2	HGB IMU+context	0.9965	0.8363	14220	50	37	189
vehicle_3	RF100 IMU	0.9763	0.8744	13202	321	27	188
vehicle_3	RF100 IMU+context	0.9826	0.8698	13288	235	28	187
vehicle_3	HGB IMU	0.9819	0.9023	13278	245	21	194
vehicle_3	HGB IMU+context	0.9896	0.9023	13382	141	21	194

Table S16: Primary PVS class recalls and confusion counts by scenario domain

Outer domain	Candidate	No-bump recall	Bump recall	TN	FP	FN	TP
scenario_1	RF100 IMU	0.9966	0.3280	15761	53	254	124
scenario_1	RF100 IMU+context	0.9979	0.2513	15780	34	283	95
scenario_1	HGB IMU	0.9977	0.3095	15777	37	261	117
scenario_1	HGB IMU+context	0.9983	0.2275	15787	27	292	86
scenario_2	RF100 IMU	0.9942	0.8478	14958	88	35	195
scenario_2	RF100 IMU+context	0.9971	0.1870	15002	44	187	43
scenario_2	HGB IMU	0.9961	0.8522	14987	59	34	196
scenario_2	HGB IMU+context	0.9983	0.2957	15021	25	162	68
scenario_3	RF100 IMU	0.9989	0.7432	11649	13	19	55
scenario_3	RF100 IMU+context	0.9973	0.7973	11631	31	15	59
scenario_3	HGB IMU	0.9992	0.7027	11653	9	22	52
scenario_3	HGB IMU+context	0.9996	0.7568	11657	5	18	56

Notes: Floating metrics use four decimal places; confusion-count cells are exact integers. Class order is 0 = no speed bump, 1 = speed bump; TN/FP/FN/TP follow this order. Macro-F1 summaries average three outer domains per axis. There are nine trips; repeated windows and axes are not independent replicates.

Independent verification reloads all 528 models, reproduces every stored prediction, reconstructs all split/cap memberships and medians, checks model parameters/classes/hashes, recomputes class recalls and confusion matrices, and recomputes every selector and endpoint. Feature verification separately recomputes 315 deterministic source win-

dows and checks all retained anchor labels/identities. Its tolerance concerns numeric representation, not annotation correctness. There are only three heldout domains per axis; no nominal row-level significance test is reported.

S12. Post hoc PVS coordinate/speed ablation

This follow-up was motivated by inspection of the v3 PVS results and was not an untouched confirmatory experiment. It keeps the parent population, labels, source-window grids, saved outer splits, uncapped test rows, capped fit rows, model settings, and training-only median imputation. It adds 48 fits (two grids × six outer folds × two classifiers × two new feature blocks) and reuses 48 independently checked sensor-only and all-context parent fits. Thus, the four blocks contribute 96 domain-level model outcomes; the full computational inventory increases to 2,286 fits, including 576 PVS fits. These are fit-fold computations over nine acquisitions, not additional independent observations. No selector, split membership, or cap index is rewritten.

The feature blocks are IMU (30 columns), IMU with speed (31), IMU with latitude/longitude (32), and IMU with latitude/longitude and speed (33). Coordinates appear in latitude, longitude order; speed is the third context column and is itself GPS-derived in this release. Thus, "GPS block" means latitude/longitude only, while the speed-bearing blocks retain GPS-derived speed. All features are float32 at model input, imputed by training-row medians, without scaling. RF100 and HGB settings match the frozen parent protocol. The contrasts are speed−IMU, GPS−IMU, GPS+speed−IMU, GPS conditional on speed (all−speed), and speed conditional on GPS (all−GPS). For each contrast, the range and number of positive, negative and zero mean effects across the eight grid × axis × classifier contexts are: speed − IMU: -0.0157 to +0.0212 across the eight prespecified grid × axis × classifier means; 4 positive, 4 negative, and 0 zero. GPS − IMU: -0.0239 to +0.0275 across the eight prespecified grid × axis × classifier means; 3 positive, 5 negative, and 0 zero. GPS+speed − IMU: -0.1034 to +0.0329 across the eight prespecified grid × axis × classifier means; 4 positive, 4 negative, and 0 zero. GPS | speed: -0.0898 to +0.0317 across the eight prespecified grid × axis × classifier means; 5 positive, 3 negative, and 0 zero. speed | GPS: -0.0795 to +0.0239 across the eight prespecified grid × axis × classifier means; 5 positive, 3 negative, and 0 zero. Tables S17–S23 retain the exact means and domain values, so sign reversals are visible. Domain folds share the same acquisitions and are descriptive, not independent replicates. A feature-block comparison changes the fitted predictive system and is not a causal intervention on location, speed, road condition, or data collection. In particular, the results do not support a universal claim that deleting GPS improves prediction. The previously declared parent two-axis estimation-error criterion remains failed.

Table S17: Domain-mean performance for the primary grid.

Axis	Model	Features	Macro-F1	Balanced accuracy
Vehicle	RF100	IMU	0.8095	0.8579
Vehicle	RF100	+speed	0.8305	0.8665
Vehicle	RF100	+GPS	0.8263	0.8642
Vehicle	RF100	+GPS+speed	0.8352	0.8695
Vehicle	HGB	IMU	0.8351	0.8768
Vehicle	HGB	+speed	0.8562	0.8839
Vehicle	HGB	+GPS	0.8625	0.8868
Vehicle	HGB	+GPS+speed	0.8679	0.8827
Scenario	RF100	IMU	0.8278	0.8181
Scenario	RF100	+speed	0.8141	0.7996
Scenario	RF100	+GPS	0.8038	0.7765
Scenario	RF100	+GPS+speed	0.7243	0.7046
Scenario	HGB	IMU	0.8341	0.8096
Scenario	HGB	+speed	0.8356	0.8027
Scenario	HGB	+GPS	0.8102	0.7607
Scenario	HGB	+GPS+speed	0.7640	0.7127

Table S18: Domain-mean performance for the nonoverlap grid.

Axis	Model	Features	Macro-F1	Balanced accuracy
Vehicle	RF100	IMU	0.7871	0.8611
Vehicle	RF100	+speed	0.7926	0.8560
Vehicle	RF100	+GPS	0.8022	0.8676
Vehicle	RF100	+GPS+speed	0.8151	0.8774
Vehicle	HGB	IMU	0.8282	0.8539
Vehicle	HGB	+speed	0.8168	0.8418
Vehicle	HGB	+GPS	0.8246	0.8498
Vehicle	HGB	+GPS+speed	0.8485	0.8587
Scenario	RF100	IMU	0.7872	0.8134
Scenario	RF100	+speed	0.7715	0.7950
Scenario	RF100	+GPS	0.7789	0.7973
Scenario	RF100	+GPS+speed	0.7862	0.7924
Scenario	HGB	IMU	0.8168	0.7914
Scenario	HGB	+speed	0.8146	0.7938
Scenario	HGB	+GPS	0.8074	0.7698
Scenario	HGB	+GPS+speed	0.7986	0.7522

Table S19: Primary vehicle-domain macro-F1 contrasts.

Model	Contrast	Mean gain	Domain 1	Domain 2	Domain 3
RF100	Speed − IMU	+0.0209	+0.0124	+0.0262	+0.0242
RF100	GPS − IMU	+0.0167	-0.0021	+0.0177	+0.0346
RF100	GPS+speed − IMU	+0.0256	-0.0057	+0.0470	+0.0355
RF100	GPS given speed	+0.0047	-0.0181	+0.0208	+0.0113
RF100	Speed given GPS	+0.0089	-0.0036	+0.0293	+0.0009
HGB	Speed − IMU	+0.0212	+0.0159	+0.0211	+0.0266
HGB	GPS − IMU	+0.0275	-0.0091	+0.0429	+0.0486
HGB	GPS+speed − IMU	+0.0329	-0.0149	+0.0555	+0.0580
HGB	GPS given speed	+0.0117	-0.0308	+0.0344	+0.0315
HGB	Speed given GPS	+0.0054	-0.0059	+0.0126	+0.0095

Table S20: Primary scenario-domain macro-F1 contrasts.

Model	Contrast	Mean gain	Domain 1	Domain 2	Domain 3
RF100	Speed − IMU	-0.0137	-0.0097	-0.0341	+0.0028
RF100	GPS − IMU	-0.0239	-0.0326	-0.0344	-0.0048
RF100	GPS+speed − IMU	-0.1034	-0.0362	-0.2462	-0.0279
RF100	GPS given speed	-0.0898	-0.0265	-0.2121	-0.0306
RF100	Speed given GPS	-0.0795	-0.0036	-0.2119	-0.0230
HGB	Speed − IMU	+0.0014	+0.0059	-0.0242	+0.0227
HGB	GPS − IMU	-0.0239	-0.0449	-0.0572	+0.0304
HGB	GPS+speed − IMU	-0.0701	-0.0451	-0.1951	+0.0298
HGB	GPS given speed	-0.0716	-0.0510	-0.1709	+0.0071
HGB	Speed given GPS	-0.0462	-0.0002	-0.1379	-0.0006

Table S21: Nonoverlap vehicle-domain macro-F1 contrasts.

Model	Contrast	Mean gain	Domain 1	Domain 2	Domain 3
RF100	Speed − IMU	+0.0055	-0.0135	+0.0261	+0.0040
RF100	GPS − IMU	+0.0151	+0.0090	+0.0231	+0.0133
RF100	GPS+speed − IMU	+0.0280	-0.0048	+0.0628	+0.0259
RF100	GPS given speed	+0.0225	+0.0087	+0.0367	+0.0219
RF100	Speed given GPS	+0.0129	-0.0138	+0.0398	+0.0126
HGB	Speed − IMU	-0.0114	-0.0135	-0.0168	-0.0041
HGB	GPS − IMU	-0.0036	-0.0410	+0.0167	+0.0135
HGB	GPS+speed − IMU	+0.0202	-0.0112	+0.0406	+0.0313
HGB	GPS given speed	+0.0317	+0.0023	+0.0573	+0.0354
HGB	Speed given GPS	+0.0239	+0.0298	+0.0239	+0.0178

Table S22: Nonoverlap scenario-domain macro-F1 contrasts.

Model	Contrast	Mean gain	Domain 1	Domain 2	Domain 3
RF100	Speed − IMU	-0.0157	-0.0123	-0.0349	+0.0000
RF100	GPS − IMU	-0.0083	-0.0161	+0.0069	-0.0158
RF100	GPS+speed − IMU	-0.0010	-0.0070	+0.0079	-0.0040
RF100	GPS given speed	+0.0147	+0.0054	+0.0428	-0.0040
RF100	Speed given GPS	+0.0073	+0.0092	+0.0010	+0.0118
HGB	Speed − IMU	-0.0021	+0.0470	-0.0089	-0.0446
HGB	GPS − IMU	-0.0094	+0.0018	-0.0063	-0.0236
HGB	GPS+speed − IMU	-0.0181	+0.0097	-0.0889	+0.0248
HGB	GPS given speed	-0.0160	-0.0374	-0.0800	+0.0693
HGB	Speed given GPS	-0.0088	+0.0079	-0.0826	+0.0484

Table S23: Vehicle-minus-scenario differences in domain-mean gains.

Grid	Model	Speed − IMU	GPS − IMU	All − IMU	GPS given speed	Speed given GPS
Primary	RF100	+0.0346	+0.0407	+0.1290	+0.0944	+0.0884
Primary	HGB	+0.0197	+0.0514	+0.1030	+0.0833	+0.0516
Nonoverlap	RF100	+0.0213	+0.0235	+0.0290	+0.0077	+0.0055
Nonoverlap	HGB	-0.0093	+0.0058	+0.0384	+0.0477	+0.0326

All contrast values are signed macro-F1 differences. Domain-level repeats and axes are not independent observations; no inferential test is implied.

Supplementary figures

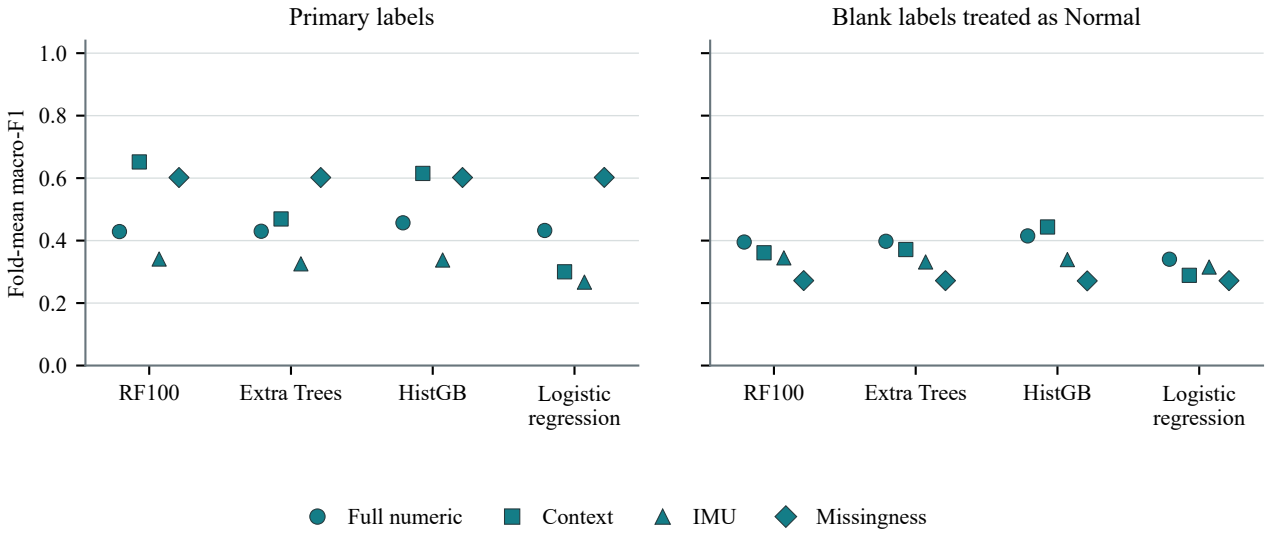


Figure S1: Recording-heldout feature comparisons for four model families. Points show fold-mean macro-F1; features are associations, not causal effects.

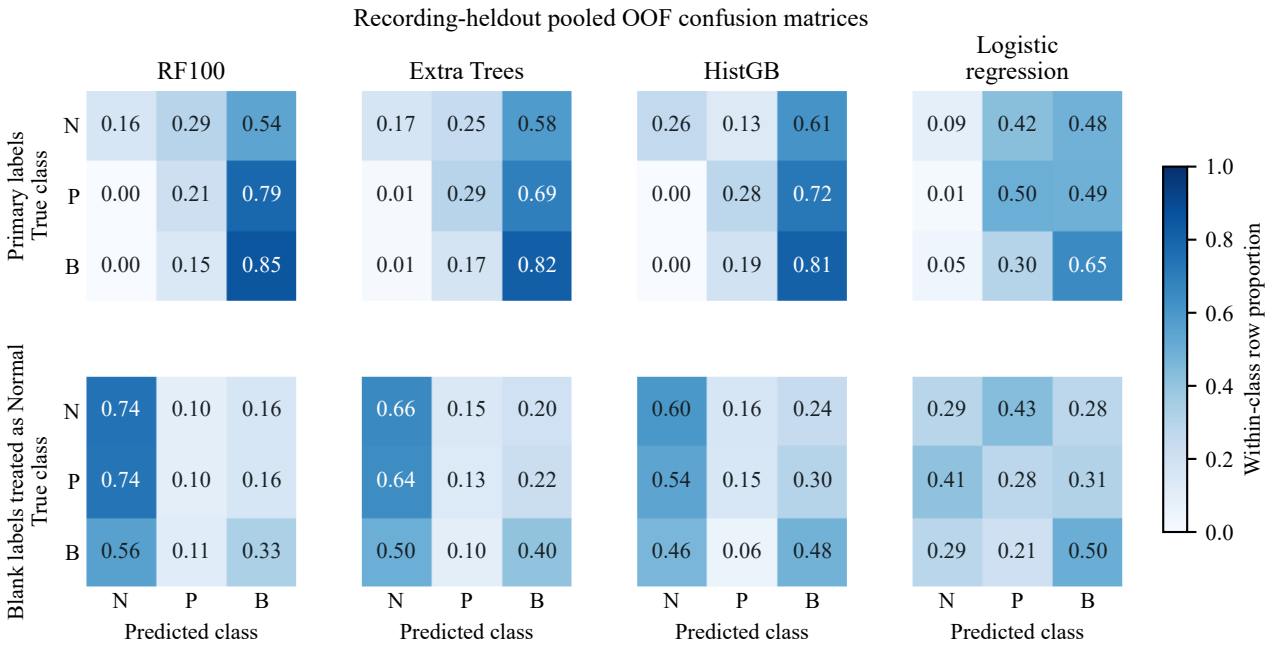


Figure S2: Full-feature recording-heldout pooled confusion matrices, normalized within true class. Diagonal entries are recall. N=Normal, P=Pothole, B=Bump.

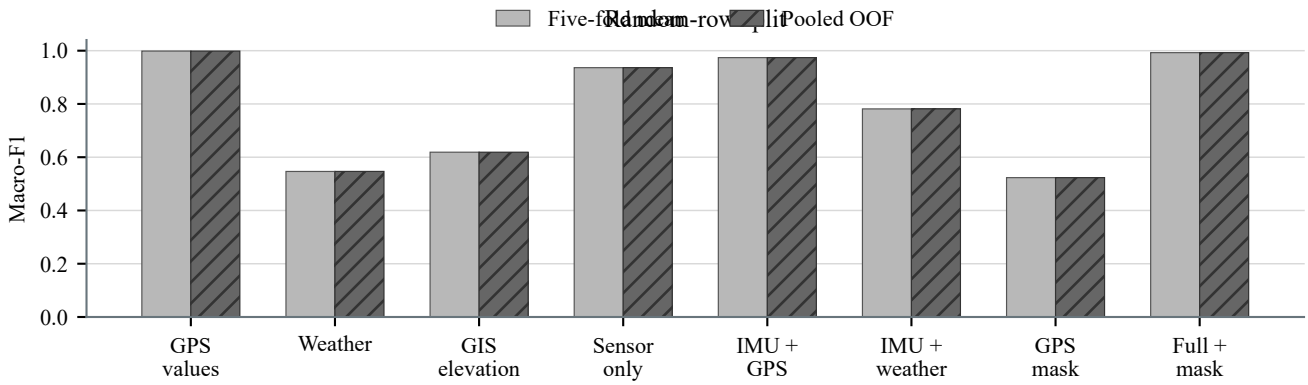


Figure S3a. RF feature ablations under the primary label policy, random-row split. Bars compare five-fold mean and pooled OOF macro-F1.

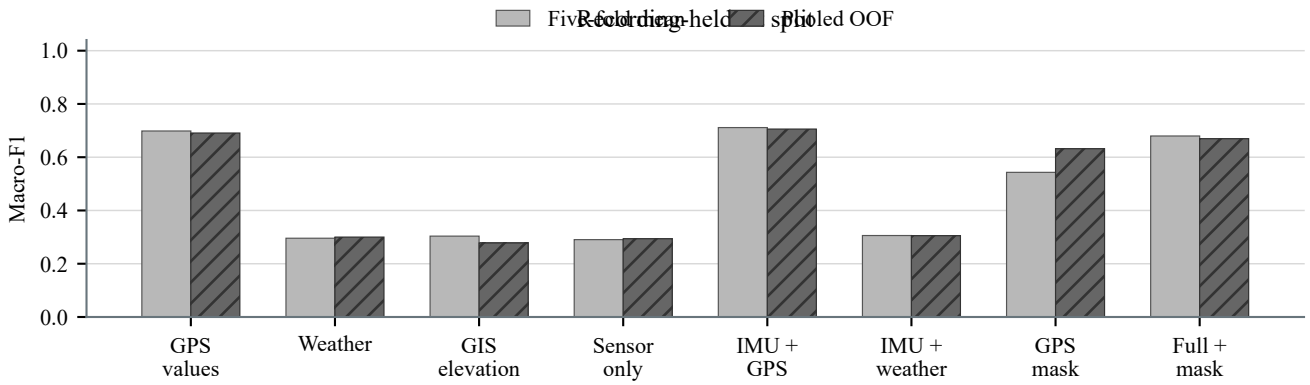


Figure S3b. RF feature ablations under the primary label policy, recording-heldout split. Bars compare five-fold mean and pooled OOF macro-F1.

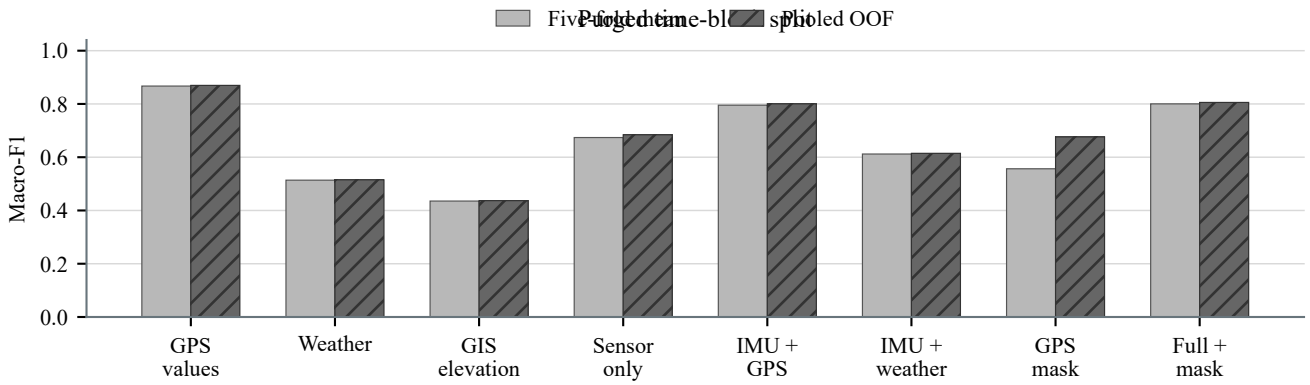


Figure S3c. RF feature ablations under the primary label policy, purged time-block split. Bars compare five-fold mean and pooled OOF macro-F1.

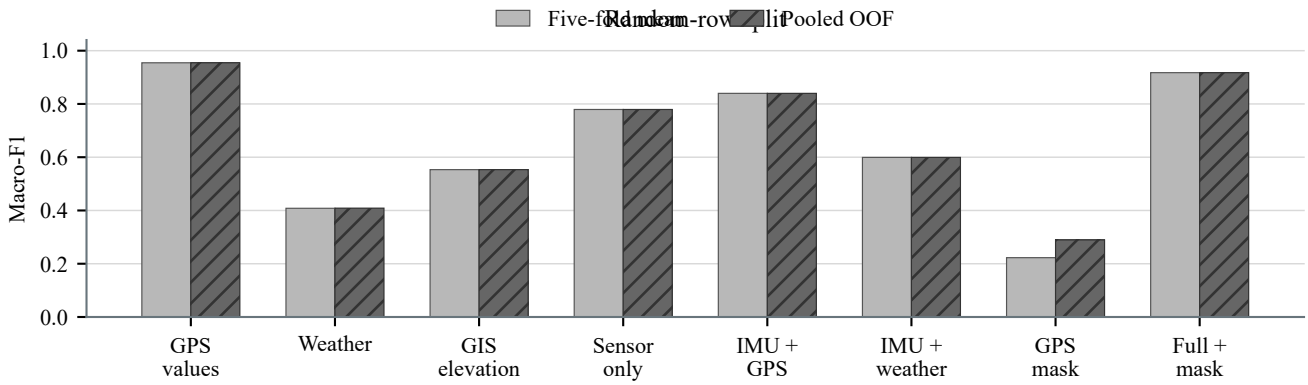


Figure S3d. RF feature ablations under the blank-as-normal sensitivity policy, random-row split. Bars compare five-fold mean and pooled OOF macro-F1.

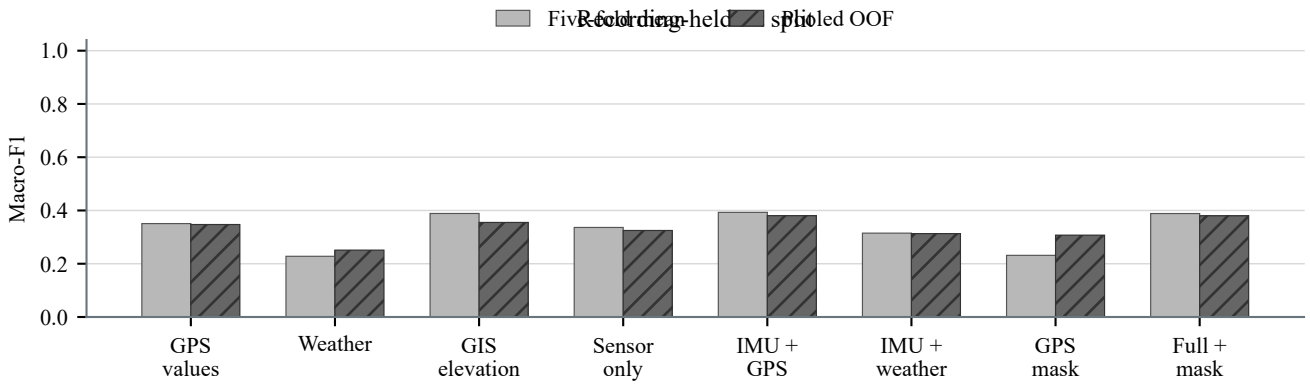


Figure S3e. RF feature ablations under the blank-as-normal sensitivity policy, recording-heldout split. Bars compare five-fold mean and pooled OOF macro-F1.

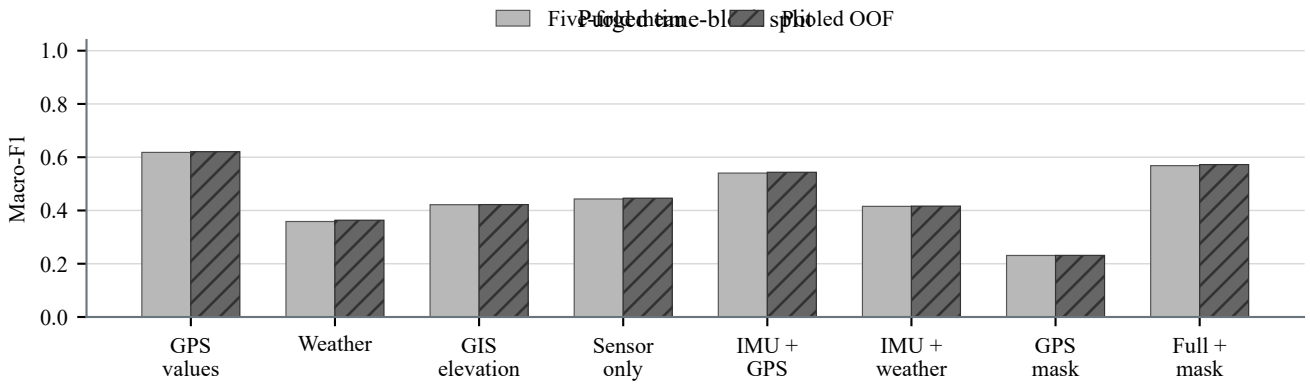


Figure S3f. RF feature ablations under the blank-as-normal sensitivity policy, purged time-block split. Bars compare five-fold mean and pooled OOF macro-F1.

S13. Common-budget PVS diagnostic

The diagnostic was fixed after inspecting prior outcomes. It adds 264 computational fits, giving 2,550 total fits including 840 PVS fits, but still only nine PVS acquisitions. V denotes vehicle/driver and S denotes scenario in the axis column; model suffix S denotes inertial sensor features and C adds coordinates and speed. Bias is outer minus inner F1. The two outer targets use different training sizes.

Table S24: Common per-class training budgets by outer domain.

Axis	Held-out domain	Class 0	Class 1
Vehicle	1	3383	53
Vehicle	2	3383	53
Vehicle	3	3568	56
Scenario	1	2917	18
Scenario	2	2917	18
Scenario	3	3763	57

Table S25: Common-budget selected-system domain outcomes.

Axis	ID	Selector	Model	Inner F1	Outer F1	Bias	Abs. error
V	1	Random	HGB/S+C	0.8684	0.8444	-0.0239	0.0239
V	1	Trip	HGB/S+C	0.8644	0.8444	-0.0200	0.0200
V	1	Matched	HGB/S	0.8383	0.8618	+0.0236	0.0236
V	2	Random	HGB/S+C	0.8549	0.8590	+0.0041	0.0041
V	2	Trip	HGB/S+C	0.8166	0.8590	+0.0425	0.0425
V	2	Matched	HGB/S+C	0.8069	0.8590	+0.0521	0.0521
V	3	Random	HGB/S+C	0.8510	0.8371	-0.0139	0.0139
V	3	Trip	HGB/S+C	0.8398	0.8371	-0.0027	0.0027
V	3	Matched	HGB/S+C	0.8415	0.8371	-0.0044	0.0044
S	1	Random	HGB/S+C	0.8960	0.6873	-0.2087	0.2087
S	1	Trip	RF/S+C	0.8919	0.7048	-0.1872	0.1872
S	1	Matched	HGB/S	0.8501	0.6842	-0.1659	0.1659
S	2	Random	HGB/S+C	0.8365	0.5588	-0.2777	0.2777
S	2	Trip	HGB/S+C	0.7730	0.5588	-0.2142	0.2142
S	2	Matched	HGB/S	0.7557	0.5447	-0.2110	0.2110
S	3	Random	HGB/S+C	0.8770	0.8787	+0.0017	0.0017
S	3	Trip	HGB/S+C	0.8517	0.8787	+0.0270	0.0270
S	3	Matched	HGB/S	0.7163	0.8565	+0.1402	0.1402

Table S26: Historical-full-budget selected-system domain outcomes.

Axis	ID	Selector	Model	Inner F1	Outer F1	Bias	Abs. error
V	1	Random	HGB/S+C	0.8684	0.8129	-0.0554	0.0554
V	1	Trip	HGB/S+C	0.8644	0.8129	-0.0515	0.0515
V	1	Matched	HGB/S	0.8383	0.8241	-0.0142	0.0142
V	2	Random	HGB/S+C	0.8549	0.9004	+0.0455	0.0455
V	2	Trip	HGB/S+C	0.8166	0.9004	+0.0838	0.0838
V	2	Matched	HGB/S+C	0.8069	0.9004	+0.0935	0.0935
V	3	Random	HGB/S+C	0.8510	0.8320	-0.0190	0.0190
V	3	Trip	HGB/S+C	0.8398	0.8320	-0.0077	0.0077
V	3	Matched	HGB/S+C	0.8415	0.8320	-0.0094	0.0094
S	1	Random	HGB/S+C	0.8960	0.7046	-0.1914	0.1914
S	1	Trip	RF/S+C	0.8919	0.7280	-0.1640	0.1640
S	1	Matched	HGB/S	0.8501	0.6949	-0.1552	0.1552
S	2	Random	HGB/S+C	0.8365	0.7800	-0.0566	0.0566
S	2	Trip	HGB/S+C	0.7730	0.7800	+0.0069	0.0069
S	2	Matched	HGB/S	0.7557	0.8688	+0.1131	0.1131
S	3	Random	HGB/S+C	0.8770	0.9113	+0.0342	0.0342
S	3	Trip	HGB/S+C	0.8517	0.9113	+0.0596	0.0596
S	3	Matched	HGB/S	0.7163	0.8865	+0.1702	0.1702

S13.1. Prospective acquisition and confirmation plan

Further evidence should come from genuinely new acquisitions across dates, route families, vehicles, and drivers, or from another independently collected event-labelled dataset with authentic acquisition identifiers. Repeated passes and physical anomalies should retain shared route and event identities. Driver and vehicle identities should be crossed where feasible. Develop any revised method, thresholds, and primary metrics on a development cohort, then freeze them before evaluation on untouched whole acquisitions. Existing PVS outer results have been inspected and cannot provide confirmation. Acquisition counts must be planned using expected effects and desired uncertainty at the acquisition or route-family level. No fixed window or fit count guarantees adequacy. The earlier dataset-search screening alone was a logistics note; the subsequently executed Mexico/Puebla subset evaluation is reported separately in S14 and does not confirm an improved method.

S14. Mexico/Puebla released-subset replication

This is an additional-source exploratory evaluation audit, not preregistration or confirmation of an improved algorithm. Source arrays, label distributions and readiness were examined before the predictive protocol was frozen. The source is the Applied Sciences publisher supplement [10]; the related Vehicles article describes the two-route outbound/return organization [11]. These publications are not counted as two independent collections. The archive's four JSON arrays contain 36,000 source positions; the reported complete population exceeds 360,000. The subset/version mapping remains unresolved. No signal statistic authenticates field provenance. No misconduct inference follows from this discrepancy or the fixed label durations.

The four source SHA256 identities, complete raw-array audit, retained infeasible v1 protocol, v 2 protocol, code and exact indices are packaged as compact evidence. No exact six-channel row or consecutive 2/5/10/25-position fragment is shared across different files; all values are finite. The six sensor channels include vibration described as derived from acceleration, so only acceleration x/y/z is used. Event annotation runs are always 6 positions (pothole) or 11(bump/braking). The first 50-position proposal retained no target events and was abandoned before any fitting. V2 fixes 10-position nonoverlapping windows at starts 0,10,...,8990 per source. Exclude any window containing braking;

otherwise retain the unique literal majority only at count 6 or more. Blank=0(non-event), pothole=1, bump=2. This release non-event is not a claim that all such physical roads are normal.

Table S27: Retained nonoverlap window counts; source files have 9,000 positions each.

Source	Non-event	Pothole	Bump	Total
Trip 1	454	48	96	598
Trip 2	444	64	93	601
Trip 3	484	67	80	631
Trip 4	516	78	80	674
Total	1,898	257	349	2,504

Outer folds hold out trips 1+2(urban) or 3+4(suburban), respectively. These route identities derive from source documentation and are not verified from released GPS. The two inner selectors are stratified random two-fold (seed 42) and leave-one-whole-trip two-fold within the single development route. A nested route-matched selector cannot be estimated here. Training is sampled without replacement to the minimum count of each class across the four inner training pools, preserving class imbalance. Budgets (no-event, pothole, bump) are (484,67,80) when holding out urban and (444,48,93) when holding out suburban. Full refits use all 1,305 or 1,199 development windows; common-budget refits use the corresponding 631 or 585 windows. The targets are reported separately. Random subsampling also changes class/row composition and does not isolate a causal sample-size effect. Exact saved memberships, rather than seed-only regeneration, are authoritative.

Each window has 36 features: mean, population SD, min, max, RMS, 25th/50th/75th percentiles and first-difference population SD for x, y, z and vector magnitude. Median imputation is fitted on actual training rows. RF uses 100 trees, depth 20, leaf minimum 5, sqrt feature sampling and balanced class weights. HGB uses 100 iterations, 31 leaves, learning rate 0.1, L2=1, balanced class weights and no early stopping. Both use seed 42; no tuning or candidate additions occur. Each selector chooses the larger pooled OOF three-class macro-F1, with RF first on exact ties. Selection is saved before outer evaluation; both selectors share the same candidate-specific outer refits. The 24 fits comprise 16 inner and 8 outer fits. All models/predictions are preserved in the workspace; the manuscript package omits raw signals and model binaries.

Table S28: Mexico task: primary full-development refit.

Held-out route	Inner selector	Chosen	Inner F1	Outer F1	Abs. error
Urban (1+2)	Random	RF	0.9922	0.9934	0.0012
Urban (1+2)	Whole-trip	RF	0.9922	0.9934	0.0012
Suburban (3+4)	Random	RF	0.9934	0.9960	0.0026
Suburban (3+4)	Whole-trip	RF	0.9950	0.9960	0.0010

F1 is macro-F1 on three fixed classes (higher is better); absolute estimation error is lower-is-better. There are two documentation-derived route families, with no population confidence interval. Both selectors share each selected outer model.

Table S29: Mexico task: separately specified common-budget outer sensitivity.

Held-out route	Inner selector	Chosen	Inner F1	Outer F1	Abs. error
Urban (1+2)	Random	RF	0.9922	0.9918	0.0004
Urban (1+2)	Whole-trip	RF	0.9922	0.9918	0.0004
Suburban (3+4)	Random	RF	0.9934	0.9960	0.0026
Suburban (3+4)	Whole-trip	RF	0.9950	0.9960	0.0010

F1 is macro-F1 on three fixed classes (higher is better); absolute estimation error is lower-is-better. There are two documentation-derived route families, with no population confidence interval. Both selectors share each selected outer model.

The mean whole-trip-minus-random selected outer-F1 difference is exactly 0 under both targets. Mean absolute-error differences are -0.0008369747 (full) and -0.0007920153 (common budget); these are descriptive means of two

route-specific pairs, not population effects. No bootstrap interval, p-value, or improvement threshold is applied. Small errors and identical selections are retained as an additional-source counterexample, not hidden because they do not reproduce PVS. Blank semantics, short windows, processed-subset provenance and only two documented route families restrict comparison with the larger RoadSens/PVS tasks. No RoadSens-trained model is transferred to Mexico.

Independent reconstruction checks all 3,600 raw-window inclusion decisions, 2,504 included label/feature records, all exact split and per-class-budget memberships, 24 model byte hashes and reloaded predictions, training imputation medians, fixed parameters, all confusion/per-class metrics, pooled candidate selection and both paired summaries. The verification code does not import the execution script or refit models. Raw reconstruction matches features within 1e-10; prediction replay uses the frozen feature values. The frozen protocol and execution SHA256 identities, preflight checksum and result hash are supplied with the evidence.

S14.1. Indonesia source expansion: label versions require clarification

The complete author repository ZIP linked to the Indonesia study [12] was obtained on 26 October 2026 and contains 60 CSV files, including concurrent phone streams, chunks and alternate exports. Readiness inspection identified six motorcycle session families and ten timestamped car-session candidates; these counts are not independent roads or necessarily complete field acquisitions. No classifier was fitted. The five base motorcycle CSVs overlap wider six-class exports and do not add acquisition units. Crucially, matching exact first 20 fields of Samsung records shows original labels agree with wider column 21 on 20,782/25,833 matched rows in M5-1 and 4,456/5,206 in M5-4. Wider column 22 also differs (22,558/25,833 and 4,693/5,206). New class codes and processed/LSTM-export fields are not documented by the base README. These wider labels are not silently accepted as ground truth. The row comparisons and source hashes are retained. Label-version clarification is required before a wider-source evaluation; mere availability of more files is insufficient. Repeated passes and overlapping road geography must remain grouped.

References

[1] A. Khandakar, D. G. Michelson, S. G. Rabbani, F. B. Shafi, M. F. Ahamed, K. R. Rahman, M. A. Rahman, M. F. Nabi, M. A. Ayari, K. Khan, and P. N. Suganthan, "RoadSens-4M: A Multimodal Smartphone & Camera Dataset for Holistic Road-way Analysis," *Scientific Data*, vol. 13, Art. no. 1036, 2026. doi: 10.1038/s41597-026-07072-y. Code: `https://github.com/rabbanishaikh/One-Million-Observations-Dataset`.

[2] A. Khandakar et al., "RoadSens-4M Dataset," Figshare, version 3, 2025 [repository citation year; v3 publication-date field: 24 February 2026]. doi: 10.6084/m9.figshare.30341143.v3. CC BY 4.0.

[3] A. Khandakar, D. G. Michelson, M. M. Mia, G. M. A. Sakeef, K. Khan, and M. A. Ayari, "CKDP-PINNFed: Dual-Layer Privacy Framework Combining Contextual K-Anonymity and Differential Privacy with Physics-Informed Federated Learning for Explainable Crowdsourced Road Anomaly Detection," *Green Energy and Intelligent Transportation*, Art. no. 100433, 2026. doi: 10.1016/j.geits.2026.100433.

[4] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017. doi: 10.1111/ecog.02881.

[5] P. Ploton, F. Mortier, M. Réjou-Méchain, N. Barbier, N. Picard, V. Rossi, C. F. Dormann, G. Cornu, G. Viennois, N. Bayol, A. Lyapustin, S. Gourlet-Fleury, and R. Pélissier, "Spatial validation reveals poor predictive performance of large-scale ecological mapping models," *Nature Communications*, vol. 11, Art. no. 4540, 2020. doi: 10.1038/s41467-020-18321-y.

[6] A. Khandakar et al., “STRIDE (Sensor Technologies for Road Insights and Driving Evaluation) Dataset,” Figshare, version 4, 2024. doi: 10.6084/m9.figshare.25460755.v4. CC BY 4.0.

[7] A. Khandakar, D. G. Michelson, M. Naznine, A. Salam, M. Nahiduzzaman, K. M. Khan, P. N. Suganthan, M. A. Ayari, H. Menouar, and J. Haider, “Harnessing Smartphone Sensors for Enhanced Road Safety: A Comprehensive Dataset and Review,” *Scientific Data*, vol. 12, Art. no. 418, 2025. doi: 10.1038/s41597-024-04193-0.

[8] J. Menegazzo and A. von Wangenheim, “Passive Vehicular Sensors (PVS) datasets,” official repository-linked release. `https://www.kaggle.com/datasets/jefmenegazzo/pvs-passive-vehicular-sensors-datasets`. Accessed 17 October 2026. CC BY-NC-ND 4.0.

[9] J. Menegazzo and A. von Wangenheim, “Speed Bump Detection Through Inertial Sensors and Deep Learning in a Multi-contextual Analysis,” *SN Computer Science*, vol. 4, Art. no. 18, 2023 [online 17 October 2022]. doi: 10.1007/s42979-022-01438-w.

[10] A. Aguilar-González and A. Medina Santiago, “CNN-Based Road Event Detection Using Multiaxial Vibration and Acceleration Signals,” *Applied Sciences*, vol.15, no.18, Art.10203, 2025. doi: 10.3390/app151810203. Publisher supplementary archive accessed 26 October 2026.

[11] A. Aguilar-González and A. Medina Santiago, “Context-Aware Road Event Detection Using Hybrid CNN–BiLSTM Networks,” *Vehicles*, vol.8, no.1, Art.4, 2026. doi: 10.3390/vehicles8010004.

[12] B. D. Setiawan et al., ‘A Machine Learning Framework for Balancing Training Sets of Sensor Sequential Data Streams, *Sensors*, vol.21, no.20, Art.6892, 2021. doi: 10.3390/s21206892. Data: `https://github.com/s-budidarma/dataset-smarphone-sensor-for-road-surface-monitoring`.